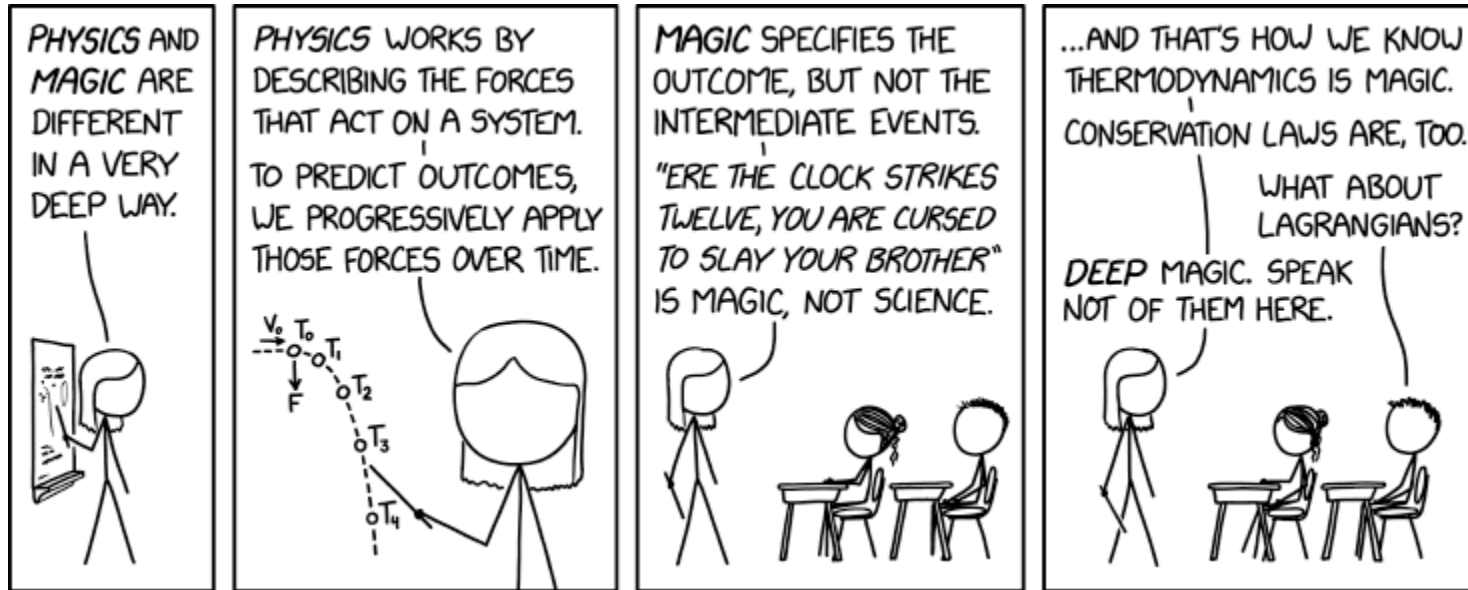




<https://xkcd.com/2904/>

Classic Mechs EP4: Sinks and Saddles and Dynamics

Classic Mechs EP4: “Eigenstuff” for dynamics

Learning objectives

- 1) Review the “Eigenstuff” we did last week for system of equations
- 2) Check out some dynamic systems in phase portrait
- 3) A little intro into Lagrange mech, where we can derive equations of motions but “ignoring” all the forces

Last week: diagonal form for the general solution

For general system $\dot{\underline{x}} = \underline{\underline{A}} \underline{x}$

$$\underline{\underline{A}} \underline{\underline{T}} = \underline{\underline{T}} \underline{\underline{D}}$$

Eigenvalues already!

T is made with columns of Eigenvectors

$$\underline{x}(t) = \underline{\underline{T}} e^{\underline{\underline{D}}t} \underline{\underline{T}}^{-1} \underline{x}(0)$$

$\underline{x}$ Single underline: Vector

$\underline{\underline{A}}$ Double underline: Matrix



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Two things before we continue:

1) Why this:

$$\underline{x}(t) = \underline{T} e^{\underline{D}t} \underline{T}^{-1} \underline{x}(0)$$

If we can:

$$\underline{x}(t) = e^{\underline{A}t} \underline{x}(0)$$

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Answer: Matrix exponential don't work like $\underline{D}$ matrix for $\underline{A}$!

Diagonalizable case [\[edit \]](#)

If a matrix is [diagonal](#):

$$A = \begin{bmatrix} a_1 & 0 & \cdots & 0 \\ 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n \end{bmatrix},$$

then its exponential can be obtained by exponentiating each entry on the main diagonal:

$$e^A = \begin{bmatrix} e^{a_1} & 0 & \cdots & 0 \\ 0 & e^{a_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^{a_n} \end{bmatrix}.$$

https://en.wikipedia.org/wiki/Matrix_exponential

Two things before we continue:

1) Why this:

$$\underline{x}(t) = \underline{T} e^{\underline{D}t} \underline{T}^{-1} \underline{x}(0)$$

If we can:

$$\underline{x}(t) = e^{\underline{A}t} \underline{x}(0)$$

Answer: Matrix exponential don't work like D matrix for A!

Two things before we continue:

2) Solving eigenvector: $\underline{\underline{A}} \underline{t}_n = \lambda_n \underline{t}_n$ For specific eigenvalue λ_n , there're a corresponding direction of eigenvector $\underline{t}_n$!

$$\left. \begin{array}{l} \underline{\underline{A}} \underline{\underline{A}} \underline{t}_n = \underline{\underline{A}} \lambda_n \underline{t}_n \rightarrow \underline{\underline{A}} \underline{\underline{A}} \underline{t}_n = \lambda_n \underline{\underline{A}} \underline{t}_n \\ \underline{\underline{A}} \underline{t}_n = \lambda_n \underline{t}_n \end{array} \right\} \underline{\underline{A}} \underline{\underline{A}} \underline{t}_n = \lambda_n \lambda_n \underline{t}_n \rightarrow \underline{\underline{A}}^2 \underline{t}_n = \lambda_n^2 \underline{t}_n$$

$\underline{\underline{A}}^n \underline{t}_n = \lambda_n^n \underline{t}_n \rightarrow$ Probability matrix predicting menu/weather $\rightarrow$ When: $\underline{\underline{A}}^n \underline{t}_{s.s.} = 1^n \underline{t}_{s.s.}$
Steady State

$\underline{t}_{s.s.}$: some times it's called principal eigenvector, and that's how Google's page ranking works.

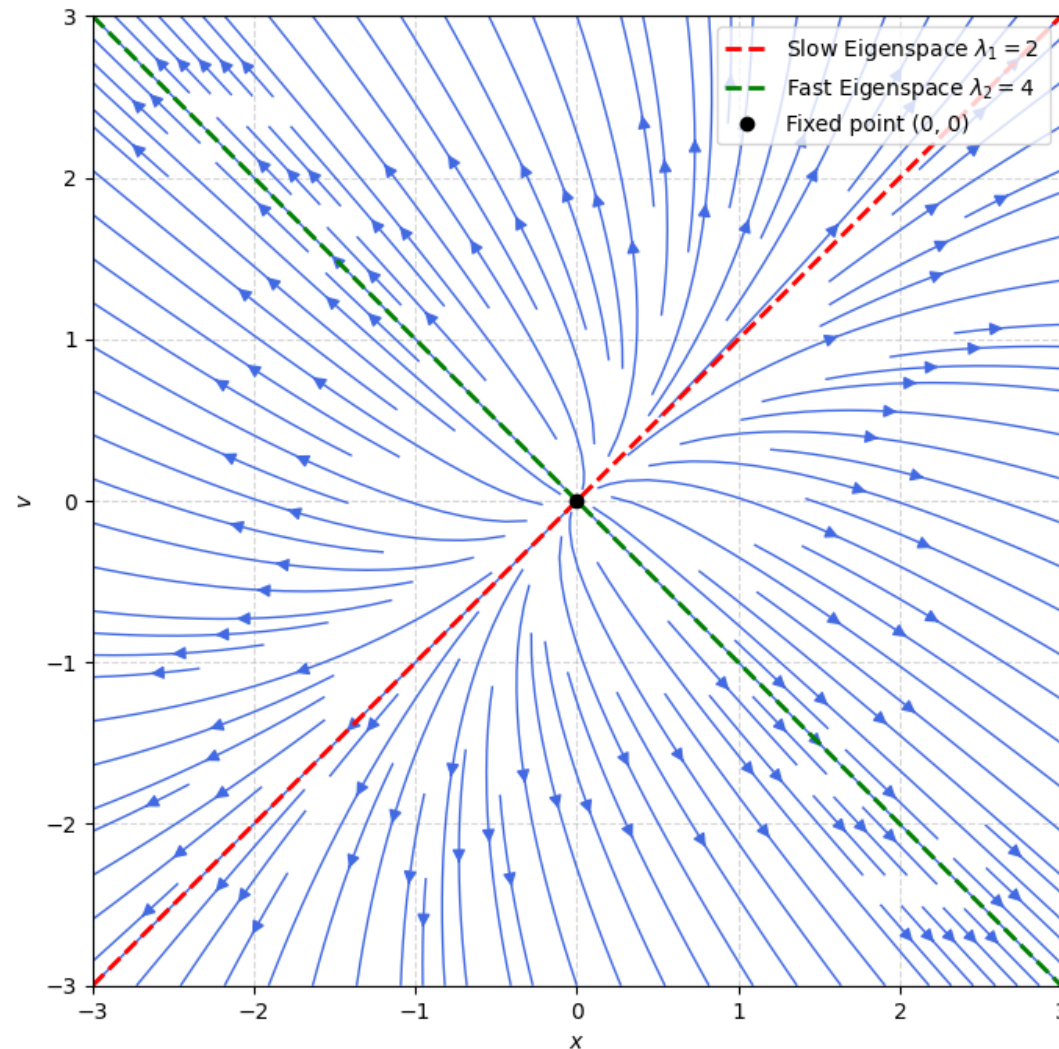
Example of unstable source:

$$\dot{\underline{x}} = \underline{A} \underline{x}$$

$$\underline{A} = \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix}$$

$$\underline{T} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\underline{D} = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$$

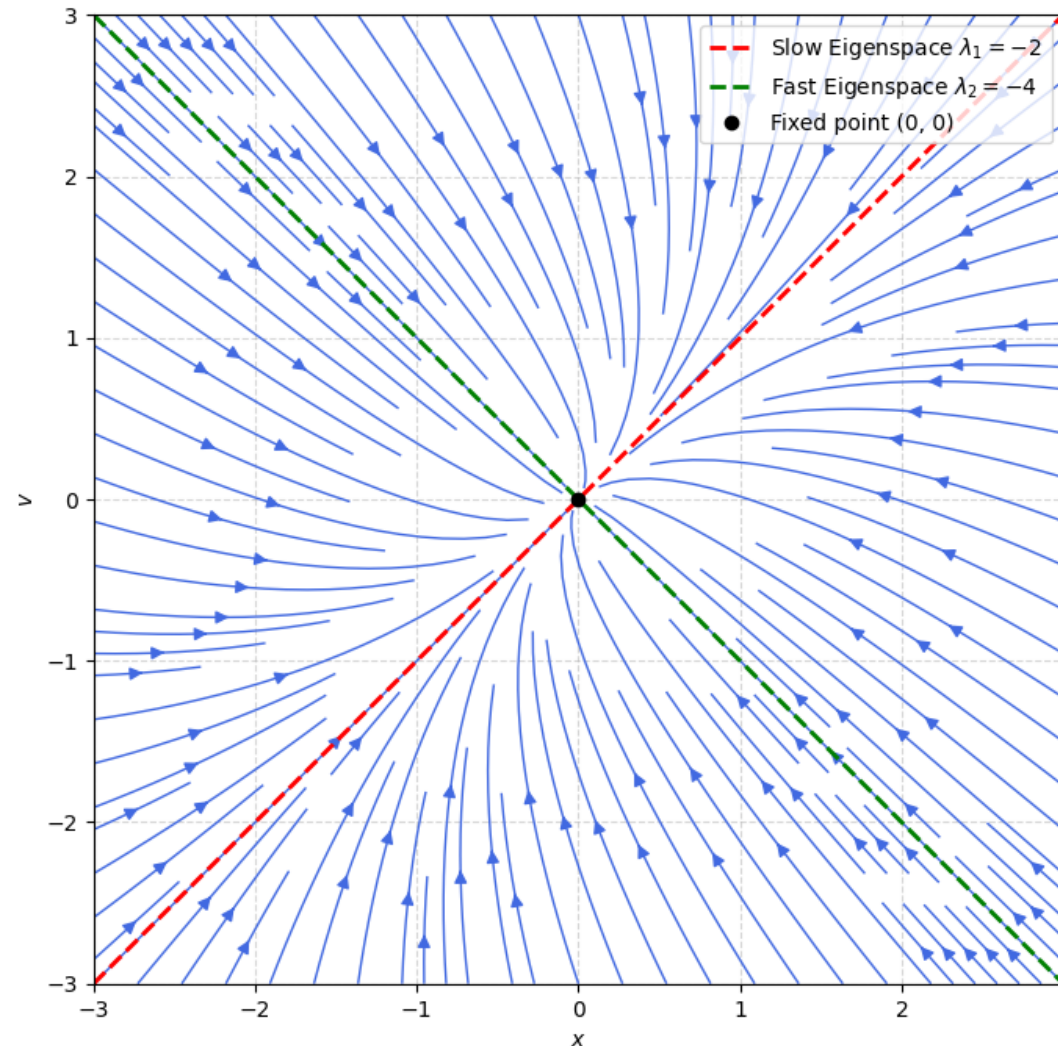


Example of stable sink:

$$\dot{\underline{x}} = \underline{A} \underline{x}$$

$$\underline{A} = -\begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix}$$

$$\underline{D} = \begin{bmatrix} -2 & 0 \\ 0 & -4 \end{bmatrix}$$



Example of saddle point:

$$\dot{\underline{x}} = \underline{A} \underline{x}$$

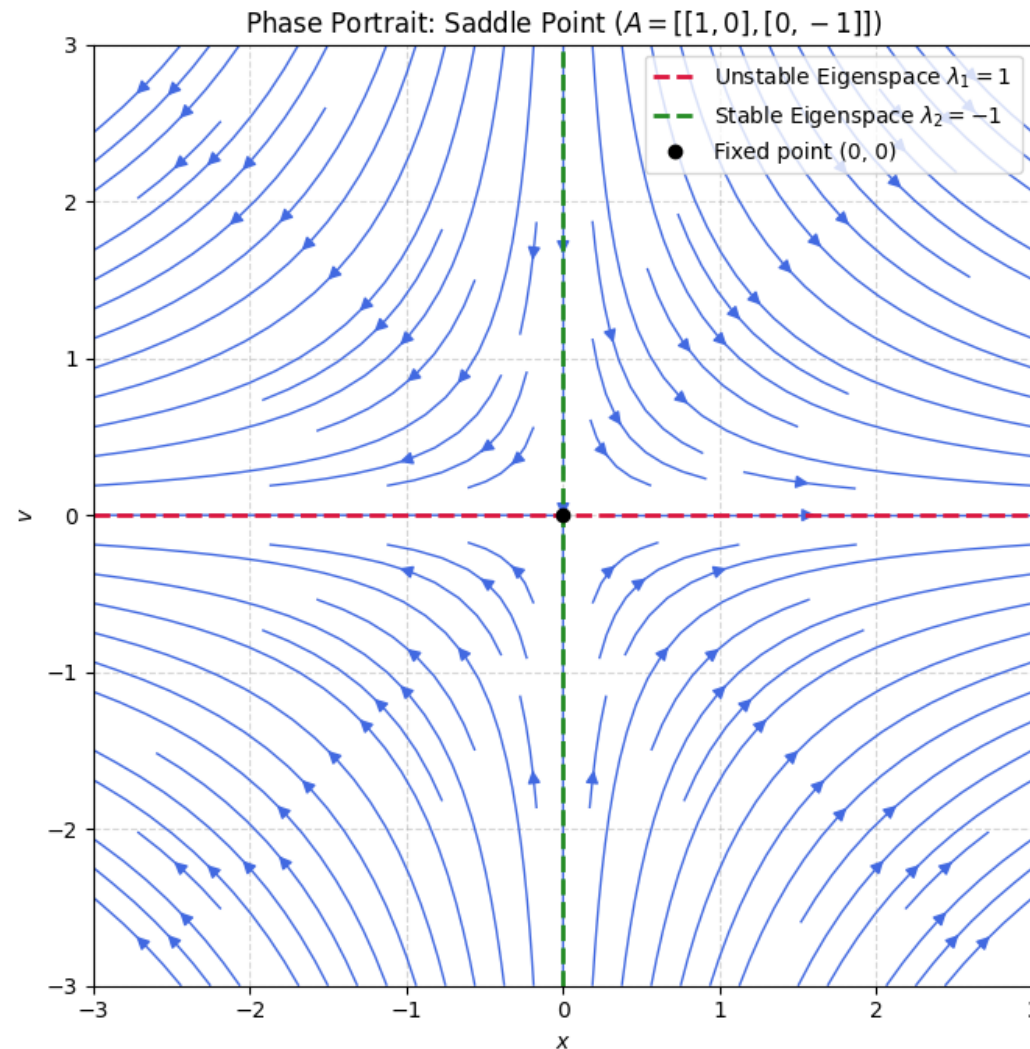
$$\underline{A} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Example of saddle point:

$$\dot{\underline{x}} = \underline{A} \underline{x}$$

$$\underline{A} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\underline{D} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



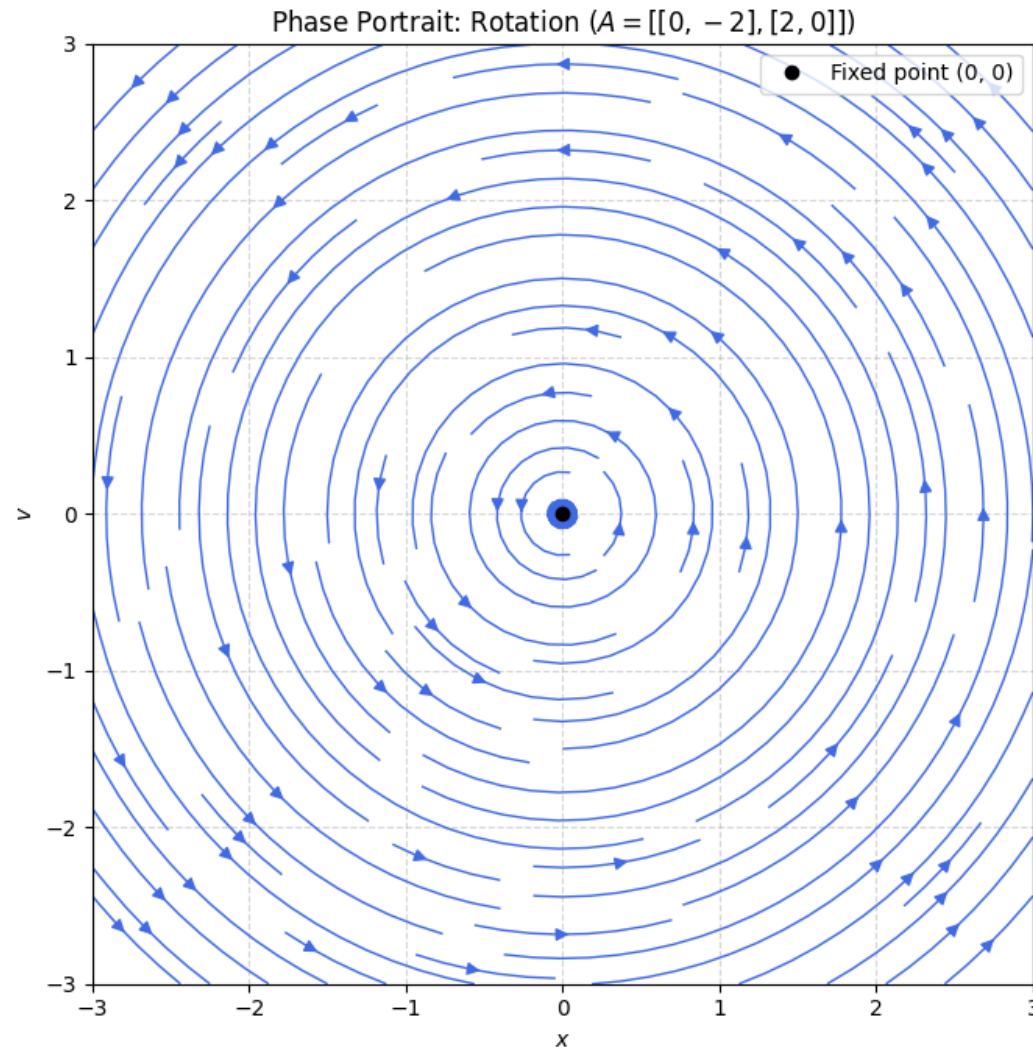
Example of Rotation/Harmonic Oscillator

$$\dot{\underline{x}} = \underline{A} \underline{x}$$

$$\underline{A} = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$$

$$\underline{D} = \begin{bmatrix} 2i & 0 \\ 0 & -2i \end{bmatrix}$$

Scaling by $2i$
along the
eigenvector
direction?



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Example of Rotation/Harmonic Oscillator

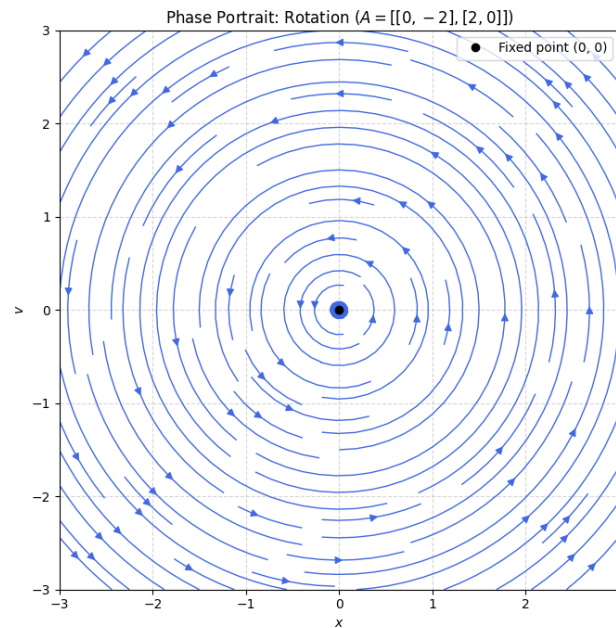
$$\dot{\underline{x}} = \underline{A} \underline{x}$$

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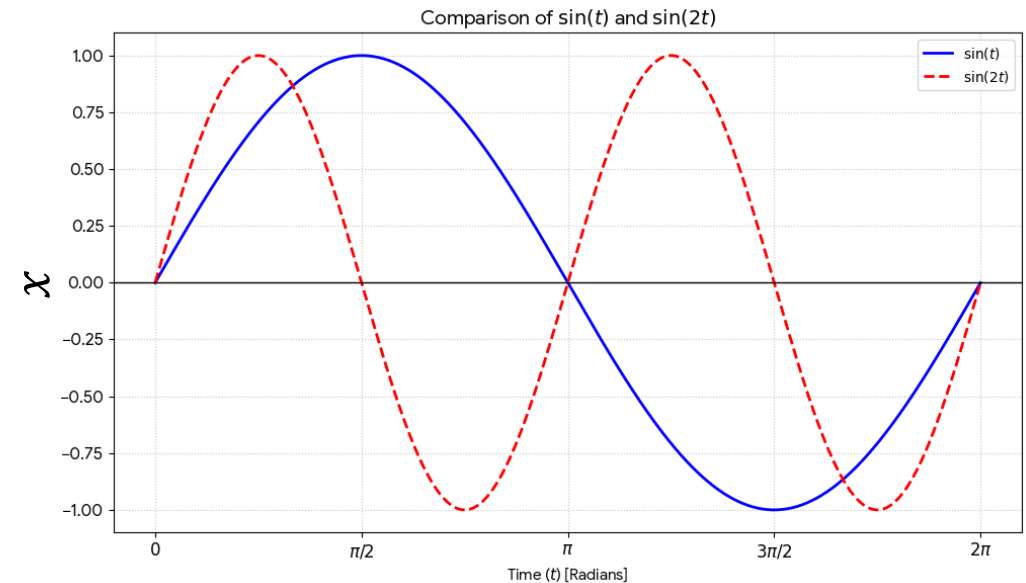
$$\underline{D} = \begin{bmatrix} 2i & 0 \\ 0 & -2i \end{bmatrix}$$

Scaling by 2i
along the
eigenvector
direction?

Can't see the scaling of 2 here.



Can tell the scaling of 2 in time domain



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Example of Rotation/Harmonic Oscillator

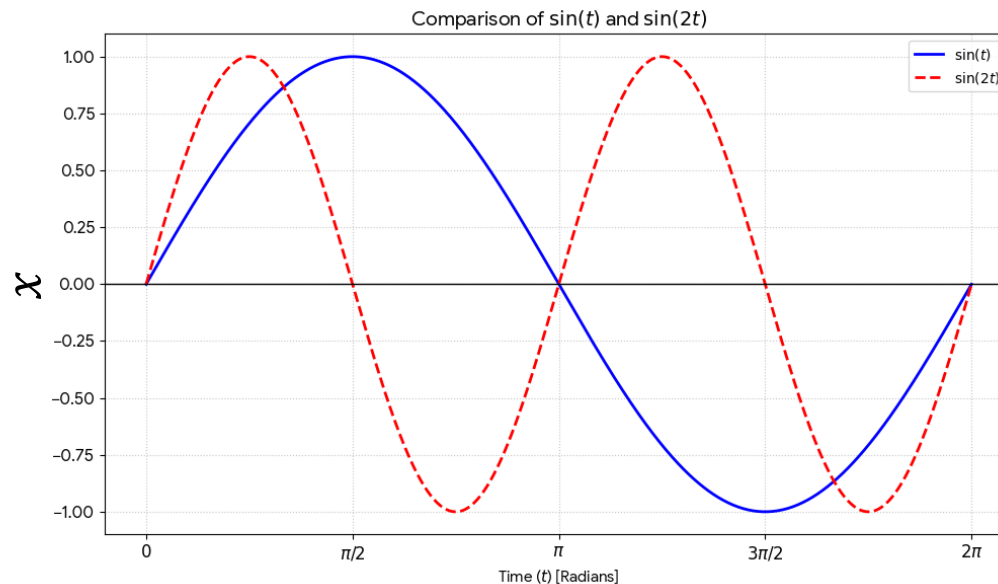
$$\dot{\underline{x}} = \underline{A} \underline{x}$$

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Scaling by 2i
along the
eigenvector
direction?

Can tell the scaling of 2 in time domain



Can see the scaling of 2 in the
frequency domain

$\mathcal{L}$: Laplace transformation

$$\mathcal{L}\{\sin(t)\} = \frac{1}{s^2 + 1}$$

$$\mathcal{L}\{\sin(2t)\} = \frac{2}{s^2 + 2^2}$$

Brief summary of eigenvalues and dynamics

$$\underline{\dot{x}} = \underline{A} \underline{x}$$

Eigenvalues	You will find:
Real: – –	Stable drain
Real: + +	Unstable source
Real: + –	Saddle
Imaginary: i, i	Rotation
Complex: – i	Stable spiral
Complex: + i	Unstable spiral

One FUN example: Sun-Jupiter gravity wells

The **A** matrix if we chose a non-inertia frame to model the dynamics of an asteroid:

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ U_{xx} & 0 & 0 & 2 \\ 0 & U_{yy} & -2 & 0 \end{bmatrix}$$

It will have four eigenvalues:

$$+i\omega, -i\omega \text{ and } +\lambda, -\lambda$$

... to induce saddle points and rotation!

